CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc (Biological Science) Examination May, 2018

BS206 Basics Genetics

Date: 15/05/18 Day: Tuesday Time: 10.00 a.m. To 10.20 a.m. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	**Choose the correct answer for the following questions.**	**10**
1.	An example of alleles is (i) AB and Tt (ii) TT and Tt (iii) T and t (iv) X and Y	
2.	Normal human eggs have (i) 22 autosomes and an X chromosome (ii) 22 autosomes and a Y chromosome (iii) 23 autosomes (iv) 46 chromosomes	
3.	DNA absorbs light at ________ nm wavelength. (i) 260 (ii) 270 (iii) 290 (iv) None of these	
4.	The four cells produced in meiosis will have (i) 2n number of chromosomes and will differ genetically from each other (ii) 2n number of chromosomes and will be genetically identical to each other (iii) n number of chromosomes and will be genetically identical to each other (iv) n number of chromosomes and will differ genetically from each other	

5.	In the F1 generation of a monohybrid cross, the phenotypic ratio would be 3:1 (True/False)	
6.	Metacentric chromosome arms are unequal in size. (True/False)	
7.	Chromosomes are best seen at the stage (i) Prophase (ii) metaphase (iii) Anaphase (iv) Telophase	
8.	A strand of DNA with the sequence A A C T T G will have a complimentary strand with the following sequence: (i) CCAGGT (ii) AACTTG (iii) TTCAAG (iv) TTGAAC	
9.	What are the repeating units of nucleic acids? (i) phosphate molecules (ii) nucleotides (iii) bases (iv) sugar molecules	
10.	A pedigree chart shows the pattern of inheritance of a specific gene. (True/False)	

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Date: 15/05/18 Day: Tuesday Time: 10.20 a.m. To 12.00 noon Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed		**10**
	Answer **any five** of the following:	
1.	Differentiate codominance and incomplete dominance.	
2.	Define the terms: (a) gene (b) chromosome	
3.	Explain DNA classification in brief.	
4	Give any four explanations for Mendels success in hybridization experiments.	
5.	Write a short note on conjugation as a mode of reproduction in bacteria.	
6.	Draw a well labeled diagram of DNA.	
SECTION - II		
Q-III Answer the following questions as directed		**15**
	Answer **any three** of the following:	
1.	What is mitosis? Explain different stages of mitosis in detail.	
2.	Compare continuous and discontinuous variation with examples.	
3.	Write 3 laws of inheritance given by Mendel. Explain the law of dominance in detail.	
4	What is probability? Discuss the sum rule and multiplication rule with examples.	
5.	Explain Griffith's Transformation Experiment.	